

MACHINE DESIGN DOCUMENT

Banking Customer Analytics & Segmentation Dashboard

A Data Analytics and Machine Learning Project

Combining Python, Scikit-learn, and Power BI

Document Type	Machine Design Document (MDD)
Domain	Banking & Financial Analytics
Technologies	Python Scikit-learn Power BI
ML Algorithm	K-Means Clustering
Dataset	3,000 Banking Customer Records
Version	1.0

Abstract

This Machine Design Document presents a comprehensive Data Analytics and Machine Learning project focused on banking customer analytics and segmentation. The project leverages a dataset of 3,000 banking customers to derive actionable financial insights through Exploratory Data Analysis (EDA) and K-Means Clustering. The end-to-end pipeline encompasses data ingestion, preprocessing, feature engineering, machine learning modeling, and interactive visualization through a six-page Power BI dashboard. The project demonstrates how data-driven methodologies can support strategic financial decision-making, including customer segmentation for targeted marketing, risk identification, and cross-selling opportunities.

Key outcomes include the identification of four distinct customer segments — VIP, High Risk, Conservative, and Low Value — alongside a suite of business insights relating to income distribution, loan behavior, deposit trends, and customer demographics. This document serves as a technical reference for both academic evaluation and industry deployment contexts.

1. Problem Statement

The banking industry generates large volumes of customer data daily. However, without systematic analysis, this data remains an underutilized asset. Financial institutions frequently struggle to:

- Identify high-value and high-risk customers within their portfolio
- Understand the relationship between customer income levels and borrowing behavior
- Develop meaningful customer segments for personalized financial products
- Make data-driven decisions regarding loan issuance and risk management

This project addresses these challenges by constructing an end-to-end analytical framework. Starting from raw customer records, the pipeline cleans and transforms the data, applies exploratory analysis to surface trends, and employs unsupervised machine learning to group customers by financial behavior. The results are surfaced through an interactive Power BI dashboard that enables non-technical stakeholders to derive insights without requiring programming expertise.

2. Objectives

The primary objectives of this project are as follows:

1	Clean and preprocess raw banking customer data to ensure analytical readiness
2	Perform comprehensive Exploratory Data Analysis (EDA) to identify trends and relationships
3	Engineer meaningful features such as Income Band, Age Group, and Total Balance
4	Apply K-Means Clustering to segment customers into actionable groups
5	Build a multi-page interactive Power BI dashboard for business stakeholders
6	Generate business insights supporting risk analysis and targeted customer strategies

3. Dataset Description

The dataset comprises 3,000 anonymized banking customer records containing demographic, financial, and account-level attributes. It was sourced as a structured CSV file and loaded into a Pandas DataFrame for analysis. The dataset initially contained 25 columns, several of which were administrative identifiers removed during preprocessing.

3.1 Dataset Overview

Total Records	3,000 customers
Total Features (raw)	25 columns
Features after cleaning	20 analytical features
File Format	CSV (.csv)
Memory Usage	586.1+ KB
Missing Values	None (all fields non-null)

3.2 Feature Categories

Customer Demographics

- Age — Customer age (integer)
- Gender (GenderId) — Encoded as 1 = Male, 2 = Female
- Nationality — Customer country of origin
- Occupation — Professional role of the customer

Financial Attributes

- Estimated Income — Annual estimated income (float64)
- Bank Loans — Outstanding loan amount (float64)
- Bank Deposits — Total bank deposit value (float64)
- Credit Card Balance — Current credit card balance (float64)
- Superannuation Savings — Retirement savings amount (float64)

Account Holdings

- Checking Accounts — Balance in checking accounts (float64)
- Saving Accounts — Balance in savings accounts (float64)
- Foreign Currency Account — Holdings in foreign currency (float64)

Business & Risk Indicators

- Risk Weighting — Risk score assigned to the customer (int64)
- Business Lending — Business-related loan exposure (float64)
- Properties Owned — Number of real estate properties owned (int64)

3.3 Dataset Preview (df.head())

The following screenshot shows the first five records of the raw dataset loaded into Pandas. The dataset spans 25 columns including identifiers, demographic fields, and financial attributes.

[4]: df.head()

[4]:

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Fee Structure	Loyalty Classification	...
0	IND81288	Raymond Mills	24	34324	06-05-2019	Anthony Torres	American	Safety Technician IV	High	Jade	...
1	IND65833	Julia Spencer	23	42205	10-12-2001	Jonathan Hawkins	African	Software Consultant	High	Jade	...
2	IND47499	Stephen Murray	27	7314	25-01-2010	Anthony Berry	European	Help Desk Operator	High	Gold	...
3	IND72498	Virginia Garza	40	34594	28-03-2019	Steve Diaz	American	Geologist II	Mid	Silver	...
4	IND60181	Melissa Sanders	46	41269	20-07-2012	Shawn Long	American	Assistant Professor	Mid	Platinum	...

5 rows × 25 columns

Figure 1: Dataset Preview — First 5 rows showing customer records with 25 columns

3.4 Dataset Structure (df.info())

The df.info() output confirms the dataset contains 3,000 non-null entries across all 25 columns, with data types including float64 (9 columns), int64 (8 columns), and object (8 columns). No missing values were detected, eliminating the need for imputation.

```
[5]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 25 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Client ID                            3000 non-null   object
1   Name                                3000 non-null   object
2   Age                                  3000 non-null   int64
3   Location ID                          3000 non-null   int64
4   Joined Bank                          3000 non-null   object
5   Banking Contact                      3000 non-null   object
6   Nationality                          3000 non-null   object
7   Occupation                           3000 non-null   object
8   Fee Structure                        3000 non-null   object
9   Loyalty Classification               3000 non-null   object
10  Estimated Income                     3000 non-null   float64
11  Superannuation Savings               3000 non-null   float64
12  Amount of Credit Cards               3000 non-null   int64
13  Credit Card Balance                  3000 non-null   float64
14  Bank Loans                           3000 non-null   float64
15  Bank Deposits                        3000 non-null   float64
16  Checking Accounts                    3000 non-null   float64
17  Saving Accounts                      3000 non-null   float64
18  Foreign Currency Account             3000 non-null   float64
19  Business Lending                     3000 non-null   float64
20  Properties Owned                     3000 non-null   int64
21  Risk Weighting                       3000 non-null   int64
22  BRId                                 3000 non-null   int64
23  GenderId                             3000 non-null   int64
24  IAIId                                3000 non-null   int64
dtypes: float64(9), int64(8), object(8)
memory usage: 586.1+ KB
```

Figure 2: Dataset Information — Column types, non-null counts, and memory usage

4. Data Preprocessing

Raw data often contains inconsistencies, redundant fields, and formatting issues that must be resolved before analysis. A systematic preprocessing pipeline was applied to this dataset to ensure data quality and analytical readiness.

4.1 Cleaning Steps

Step 1: Column Name Standardization

Column names containing spaces were stripped and reformatted using underscore notation to ensure compatibility with Python and Power BI. For example, 'Bank Loans' became 'Bank_Loans'. This was achieved using:

```
df.columns = df.columns.str.strip().str.replace(' ', '_')
```

Step 2: Dropping Irrelevant Columns

Five administrative columns were dropped as they contained no analytical value: Client_ID, Name, Banking_Contact, BRId, and IAId. Retaining these would not contribute to any model or visualization, and removing them reduces noise in the data.

Step 3: Date Type Conversion

The Joined_Bank column was originally stored as a string object. It was converted to a proper datetime format with day-first parsing to enable time-based analysis (e.g., deposit trends over time) in Power BI.

1.Cleaning Column Names

```
[9]: df.columns = df.columns.str.strip().str.replace(" ", "_")
```

2.Dropping Unnecessary Columns

```
[13]: df.drop([
    'Client_ID',
    'Name',
    'Banking_Contact',
    'BRId',
    'IAId'
],axis=1,inplace=True)
```

3.Converting Date Properly

```
[14]: df['Joined_Bank']=pd.to_datetime(df['Joined_Bank'],dayfirst=True)
```

Creating Income Band

```
[15]: df['Income_Band']=pd.cut(
    df['Estimated_Income'],
    bins=[0,50000,150000,1000000],
    labels=['Low','Medium','High']
)
```

Creating Age Group Band

```
[17]: df['Age_Group']=pd.cut(
    df['Age'],
    bins=[0,30,60,100],
    labels=['Yong','Middle','Senior']
)
```

Figure 3: Data Cleaning Code — Column standardization, dropping irrelevant columns, and date conversion

4.2 Feature Engineering

Three derived features were engineered to enrich the dataset with business-relevant groupings and aggregates:

Feature	Method	Description / Bins
Income_Band	pd.cut()	Low: 0–50K Medium: 50K–150K High: 150K–1M
Age_Group	pd.cut()	Young: 0–30 Middle: 30–60 Senior: 60–100
Total_Balance	Arithmetic sum	Bank_Deposits + Saving_Accounts + Checking_Accounts

Creating Income Band

```
] : df['Income_Band']=pd.cut(  
    df['Estimated_Income'],  
    bins=[0,50000,150000,1000000],  
    labels=['Low','Medium','High']  
    )
```

Figure 4: Feature Engineering Code — Income Band creation using `pd.cut()`

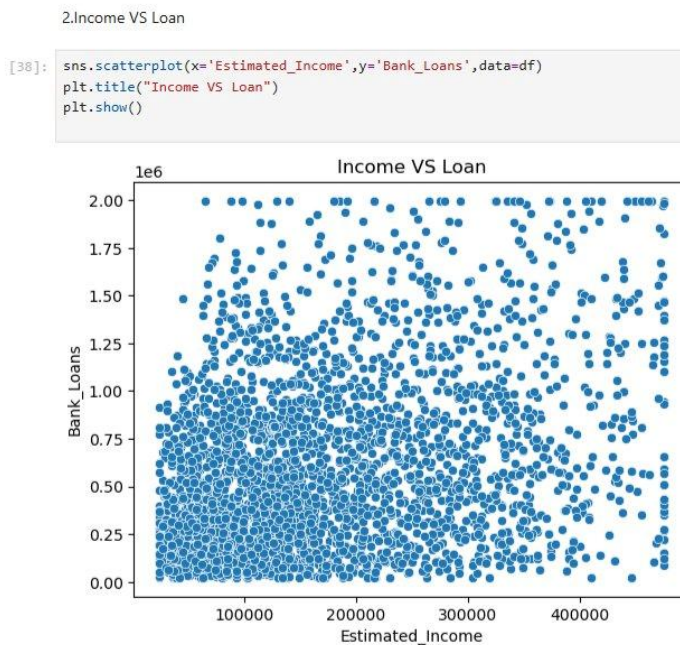
The `Income_Band` feature categorizes customers into three financial tiers, enabling comparative analysis of loan and deposit behavior. The `Age_Group` segmentation supports demographic profiling. `Total_Balance` provides a single holistic measure of a customer's overall account wealth, which proved to be the strongest predictor in the clustering model.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the distribution of key variables, identify correlations, and surface preliminary insights before applying machine learning. Five key visualizations were produced using Seaborn and Matplotlib.

5.1 Income vs. Loan Amount (Scatter Plot)

A scatter plot was constructed plotting Estimated_Income on the X-axis against Bank_Loans on the Y-axis for all 3,000 customers. The plot reveals a wide dispersion of data points without a clear linear trend, indicating that income alone is not a reliable predictor of loan size. Customers across all income levels carry varying loan amounts, suggesting that individual circumstances, risk appetite, and credit history play significant roles in loan uptake.



Here we can observe that Loan amount is weakly related to income

Figure 5: Scatter Plot — Income vs. Bank Loans (weak linear relationship)

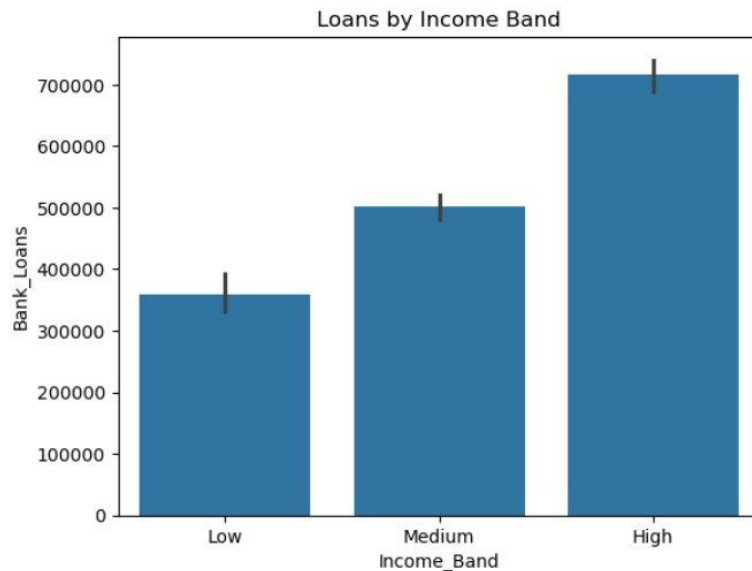
Business Insight: Loan underwriting decisions should not rely solely on income metrics; behavioral and credit history indicators provide stronger signals.

5.2 Loans by Income Band (Bar Chart)

A grouped bar chart was produced to compare average loan amounts across the three income bands: Low, Medium, and High. The chart shows a clear upward progression, with High-income customers holding the largest average loan balances (~700K), followed by Medium (~500K) and Low (~360K). However, this does not necessarily translate to higher risk — high-income customers have greater repayment capacity.

3.Loan by Income Band

```
[39]: sns.barplot(x='Income_Band',y='Bank_Loans',data=df)
plt.title("Loans by Income Band")
plt.show()
```



Here we can see that People with High income band takes more loan but also has high income so they have lower risk Low-income customers may represent higher risk because their repayment capacity is lower relative to their loan amounts.

Figure 6: Bar Chart — Loans by Income Band (High-income customers take larger loans)

Business Insight: Low-income customers exhibit disproportionately high loan-to-income ratios, representing elevated credit risk. Risk models should apply stricter thresholds for this segment.

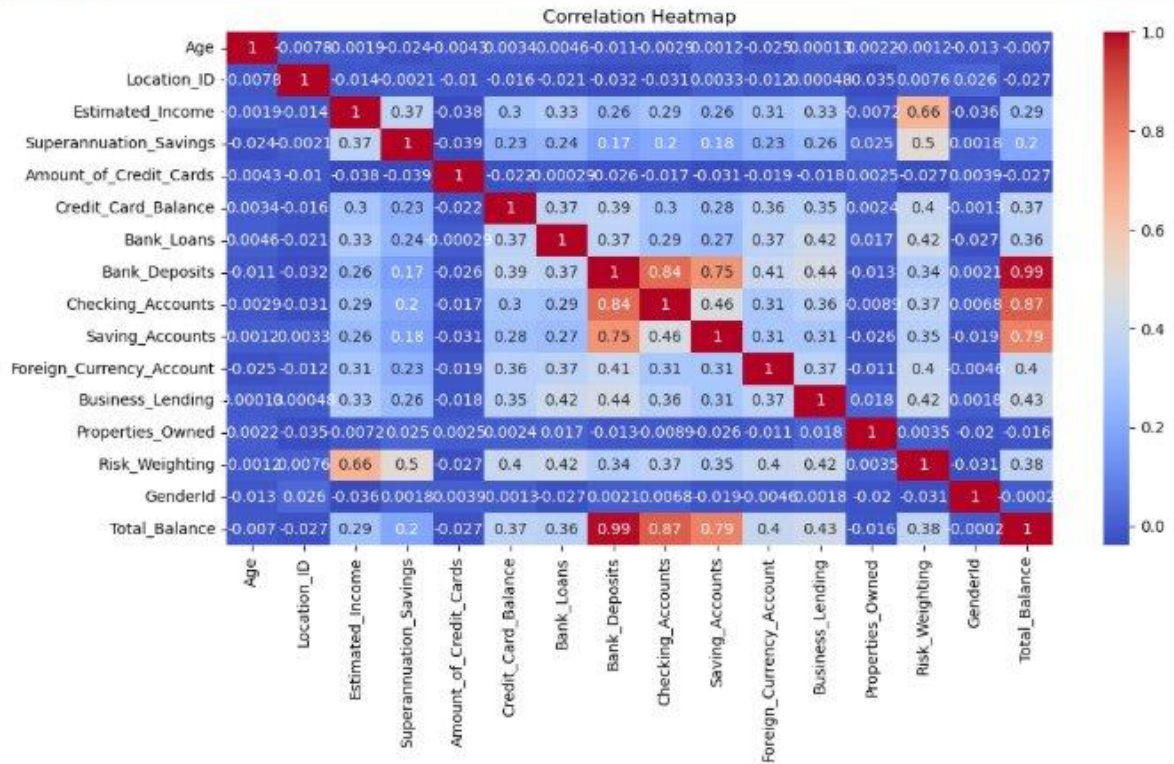
5.3 Correlation Heatmap

A Pearson correlation heatmap was generated for all numerical variables. The heatmap reveals several significant relationships:

- Total_Balance shows near-perfect correlation (0.99) with Bank_Deposits — confirming deposits as the primary driver of overall customer wealth
- Bank_Deposits and Checking_Accounts are strongly correlated (0.84), indicating customers with high deposits also maintain large checking balances
- Estimated_Income shows moderate positive correlations with Bank_Deposits (0.37), Bank_Loans (0.33), and Business_Lending (0.33)
- Age and Gender show minimal correlations with financial variables, suggesting behavior is driven more by financial profile than demographics
- Risk_Weighting correlates positively with both income (0.66) and loan amounts (0.5), confirming its validity as a risk indicator

4. Correlation Heatmap (Relationship Between income, loans, deposits)

```
[41]: plt.figure(figsize=(12,6))
sns.heatmap(df.corr(numeric_only=True),annot=True,cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



Here we can observe that correlation analysis shows that total balance is strongly influenced by deposits and account holdings. Income has a moderate impact on financial activity, while risk weighting is significantly associated with both income and loan exposure. Additionally, strong correlations among banking products indicate cross-selling opportunities, while demographic factors like age and gender have minimal impact.

Figure 7: Correlation Heatmap — Relationships between all numerical features

Business Insight: Bank Deposits is the most powerful indicator of customer value. Cross-selling strategies should focus on customers with high deposit balances who have yet to take loans.

6. Machine Learning Approach

An unsupervised machine learning approach was adopted using the K-Means Clustering algorithm to segment customers into behaviorally distinct groups. This methodology is well-suited for scenarios where ground-truth labels are unavailable and the goal is pattern discovery within financial data.

6.1 Feature Selection

Four numerical features were selected for clustering based on their analytical relevance and correlation strengths identified during EDA:

- `Estimated_Income` — represents earning capacity
- `Bank_Loans` — represents borrowing behavior
- `Bank_Deposits` — the strongest financial indicator
- `Total_Balance` — composite wealth metric

These features capture both the income side and the asset/liability side of a customer's financial profile, providing a holistic view for segmentation.

6.2 Data Scaling

Prior to clustering, all four features were standardized using `StandardScaler` from Scikit-learn. This is a critical preprocessing step for K-Means because the algorithm is distance-based — unscaled features with larger absolute values (e.g., `Bank_Loans` in millions vs. `Income` in hundreds of thousands) would dominate the distance calculations and bias the clusters.

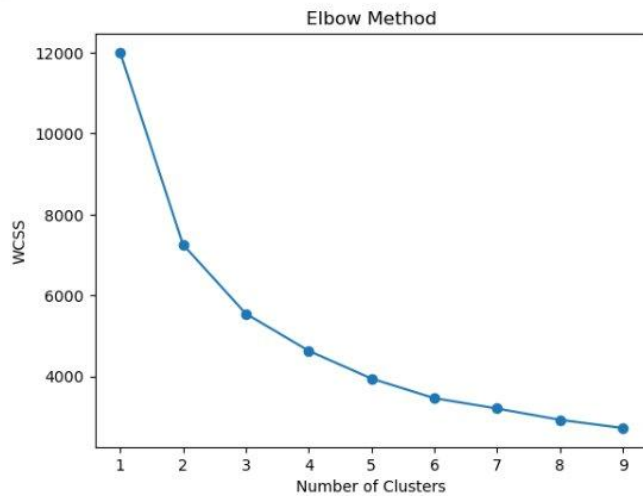
6.3 Elbow Method — Optimal k Selection

The Elbow Method was employed to determine the optimal number of clusters. K-Means was run iteratively for $k = 1$ to 9, recording the Within-Cluster Sum of Squares (WCSS) — also known as inertia — at each step. The curve produced exhibits a characteristic 'elbow' shape, with sharp WCSS reduction from $k=1$ to $k=3$, followed by diminishing returns beyond $k=4$.

```
[50]: wcss = []

for i in range(1, 10):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(scaled_data)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,10), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



Here we can observe that The curve drops sharply from $k = 1$ to $k = 3$. After $k = 4$, the decrease becomes gradual (flattens). The elbow point is observed at $k = 4$, indicating that 4 clusters provide the best balance between minimizing WCSS and avoiding overfitting.

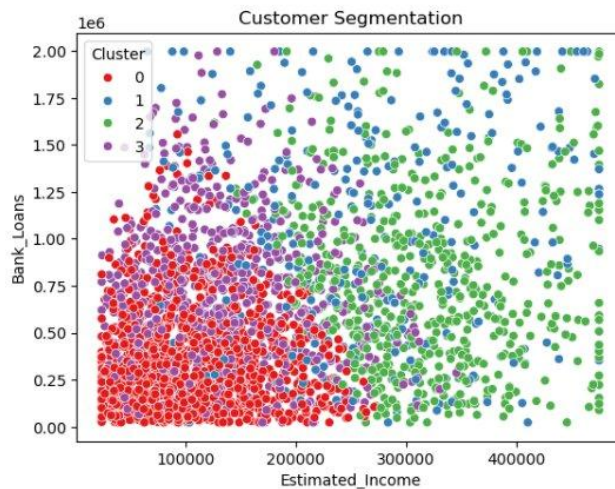
Figure 8: Elbow Method — WCSS vs. Number of Clusters (elbow at $k = 4$)

The elbow point is clearly observed at $k = 4$, indicating that four clusters provide the optimal balance between model complexity and explanatory power. Using fewer clusters would over-generalize, while more clusters would yield marginal improvements at the cost of interpretability.

6.4 Cluster Visualization

After fitting K-Means with $k=4$ on the standardized data, cluster labels were assigned back to the original DataFrame. A scatter plot was generated with Estimated_Income on the X-axis and Bank_Loans on the Y-axis, colored by cluster assignment, revealing four visually distinct customer groups.

```
[52]: sns.scatterplot(
      x='Estimated_Income',
      y='Bank_Loans',
      hue='Cluster',
      data=df,
      palette='Set1'
    )
plt.title("Customer Segmentation")
plt.show()
```



Here we can observe that
 Cluster 0 (Red): Low income, low loans → Low value customers
 Cluster 1 (Blue): Medium-high income, high loans → Active borrowers
 Cluster 2 (Green): High income, moderate loans → Premium customers
 Cluster 3 (Purple): Mixed income & loans → Balanced/average customers

Customers are segmented into four groups based on income and loan behavior, helping identify low-value, premium, and high-loan customer segments for targeted strategies

Figure 9: Customer Segmentation Scatter Plot — Four clusters distinguished by income and loan behavior

Cluster interpretations based on visual analysis:

- Cluster 0 (Red): Low income, low loans — Low-value customers with minimal financial engagement
- Cluster 1 (Blue): Medium-to-high income, high loans — Active borrowers with significant loan exposure
- Cluster 2 (Green): High income, moderate loans — Premium customers with strong earning capacity
- Cluster 3 (Purple): Mixed income and loans — Balanced/average customers with moderate financial profiles

6.5 Cluster Summary Statistics

The cluster means were computed across the four key features to quantitatively characterize each segment:

Step 4: Understanding each cluster

```
[53]: df.groupby('Cluster')[[
      'Estimated_Income',
      'Bank_Loans',
      'Bank_Deposits',
      'Total_Balance'
    ]].mean()
```

	Estimated_Income	Bank_Loans	Bank_Deposits	Total_Balance
Cluster				
0	103816.561137	3.450670e+05	2.722025e+05	5.510529e+05
1	262156.672816	1.062330e+06	2.091485e+06	3.586497e+06
2	317790.396403	7.688092e+05	4.837662e+05	9.676834e+05
3	135566.844331	7.132164e+05	1.004808e+06	1.777104e+06

Figure 10: Cluster Summary Table — Mean values of key features per cluster

Cluster	Avg. Income	Avg. Loans	Avg. Deposits	Segment Label
0	103,817	345,094	272,203	Low Value
1	262,157	1,062,330	2,091,485	VIP / High Deposits
2	317,790	768,809	483,766	Conservative / Premium
3	135,567	713,200	1,004,808	High Risk / Borrowers

6.6 Business Segment Mapping

For dashboard and business reporting purposes, the four clusters were mapped to descriptive segment labels aligned with banking terminology:

Segment	Cluster	Characteristics
Low Value	Cluster 0 (Red)	Low income (~104K), low loans (~345K), low deposits (~272K). Minimal banking engagement.
VIP	Cluster 1 (Blue)	High deposits (~2.09M) relative to peers. High value customers with significant assets.
Conservative	Cluster 2 (Green)	Highest income (~318K), moderate loans relative to income. Lower risk profile.
High Risk	Cluster 3 (Purple)	Moderate income (~136K) but high loans (~713K). Elevated loan-to-income ratio.

7. Power BI Dashboard Design

A comprehensive six-page interactive dashboard was developed in Microsoft Power BI Desktop. The dashboard employs a dark navy-and-blue corporate theme consistent with financial industry standards. Each page is accessible via a persistent left-side navigation panel with icons for Overview, Loans, Deposits, Customers, Segments, and Settings.

7.1 Overview Dashboard

The Overview page serves as the executive summary of the entire analytics suite. It presents four high-level KPI cards at the top — Total Customers (3K), Total Loans (1.77bn), Total Deposits (2.00bn), and Total Balance (3.68bn). Below the KPIs, four visualizations provide a comprehensive snapshot: an Income vs. Loans scatter plot colored by customer segment, a Deposits Growth Over Time line chart revealing a sharp uptick near 2020, a Customer Segmentation donut chart (46.43% Low Value, 22.87% Conservative, 20.53% High Risk, 10.17% VIP), and a Customer Age Distribution bar chart. Global slicers for Income_Band, Gender, and Segment are positioned in the top-right area and apply across all charts on the page.



Figure 11: Power BI Overview Dashboard — Executive KPIs, segment overview, and deposit trends

7.2 Loans Analysis Dashboard

The Loans page is dedicated to loan-specific analytics. Four KPI cards display Total Loan Amount (1.77bn), Average Loan Amount (589.18K), Loan-to-Income Ratio (3.44), and High Risk Customers (686). The page contains a Loans by Income Band horizontal bar chart (High-income customers hold ~1.0bn in loans), an Income vs. Loans scatter plot segmented by customer group, and a Loan by

Segmentation donut chart that reveals relatively even loan distribution across the four segments (High Risk: 27.68%, Low Value: 27.19%, Conservative: 26.79%, VIP: 18.33%).



Figure 12: Power BI Loans Dashboard — Loan KPIs, income-band distribution, and segment breakdown

7.3 Deposits Analysis Dashboard

The Deposits page provides a detailed view of deposit behavior across customer segments. KPI cards show Total Deposits (2.00bn), Average Deposits per Customer (668.13K), Total Savings (698.73M), and Total Checking (963.28M). A Deposits Trends line chart plots aggregate deposits by year (1995–2021), revealing a significant spike around 2020. A Deposits by Income Band chart confirms high-income customers hold approximately 1.0bn in deposits. The Deposits by Segment donut chart shows VIP customers account for 34.39% of all deposits, despite representing only 10.17% of total customers — confirming their disproportionate value. An Account Type Breakdown stacked bar chart shows the composition of Checking, Savings, and Foreign Currency accounts per segment.



Figure 13: Power BI Deposits Dashboard — Deposit trends, segment distribution, and account breakdown

7.4 Customer Analysis Dashboard

The Customer page focuses on demographic profiling. Four KPI cards present Total Customers (3K), Average Age (51.04), Average Income (171.11K), and Average Balance (1.23M). A Customer Age Distribution chart reveals the Middle age group (30–60) is the largest segment, followed by Senior (60+) and Young (under 30). A Gender Split donut chart shows near-equal distribution (Male: 50.4%, Female: 49.6%). A Customer Income Distribution horizontal bar chart shows High and Medium income bands dominate the portfolio. The Top Occupations bar chart identifies roles such as Web Designer II and Web Developer as the most common occupations among customers.

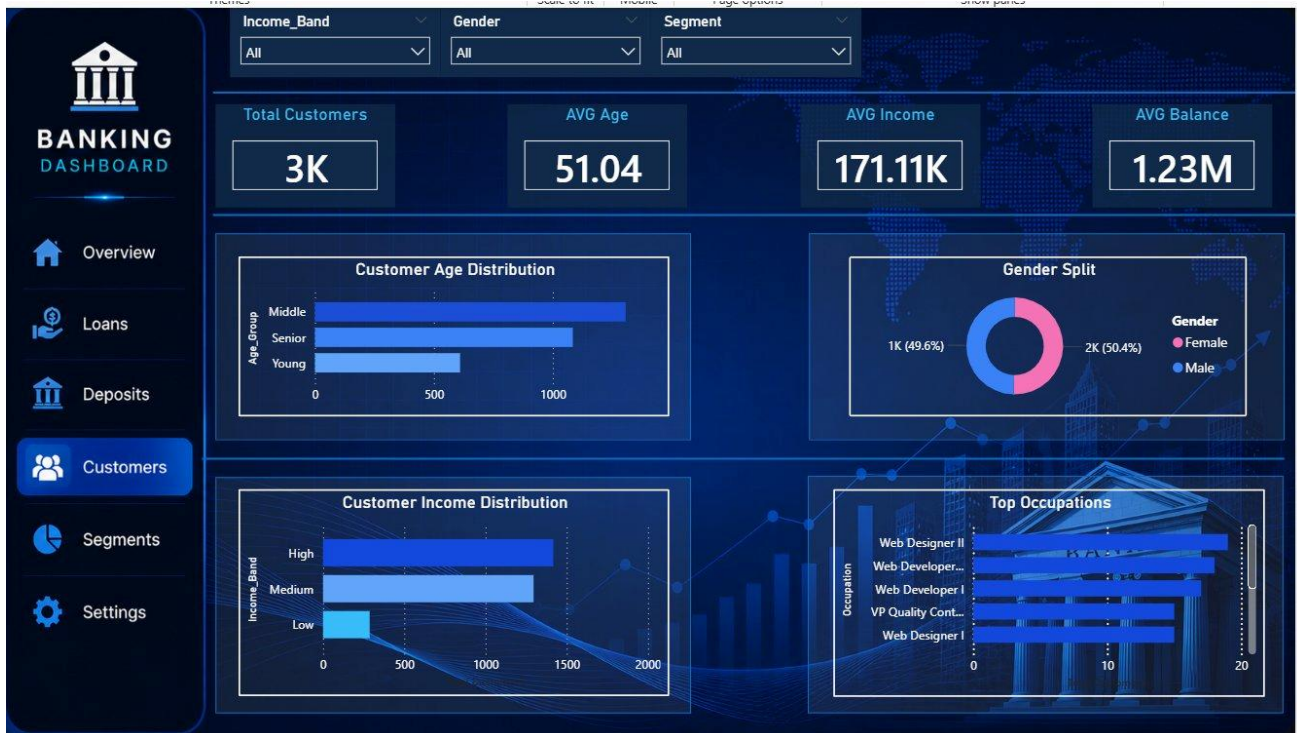


Figure 14: Power BI Customer Dashboard — Demographics, income distribution, gender split, and occupations

7.5 Segmentation Dashboard

The Segmentation page is the ML-integrated view, presenting the four K-Means clusters as business segments. Four summary KPI cards display Total Segments (4), Total Customers (3K), Average Balance (1.23M), and High Value Customers (305). A Customer Segmentation donut chart presents the segment distribution, while a Customer Value by Segment horizontal bar chart reveals that the High Risk and VIP segments hold the highest total balances. An Income vs. Loans clustered column chart compares income and loan totals per segment. An Account Type Breakdown stacked bar chart (Loans vs. Deposits by segment) supports cross-selling strategy identification.



Figure 15: Power BI Segmentation Dashboard — Cluster-based KPIs, segment value, and income-loan analysis

7.6 Dashboard Settings Panel

The Settings page centralizes all interactive filter controls and provides a reference guide. Four dropdown slicers (Income_Band, Gender, Segment, Age_Group) allow cross-page filtering. A date range slider controls the Joined_Bank filter, spanning from 03-01-1995 to 31-12-2021. A 'Reset Filters' button clears all active selections to return the dashboard to its default state. A descriptive text panel at the bottom summarizes the dashboard's capabilities and the technology stack (Python + Power BI + Machine Learning).

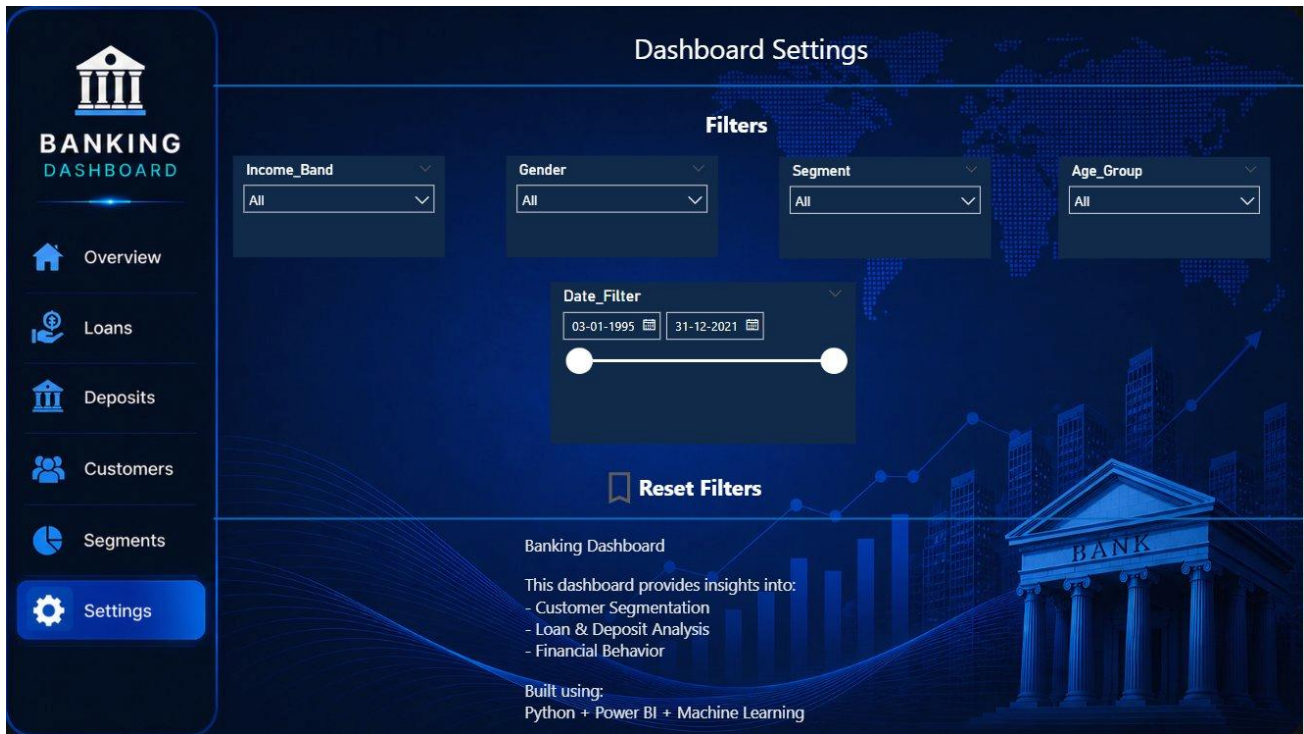


Figure 16: Power BI Settings Panel — Centralized filters, date range, and reset functionality

8. System Architecture

The project follows a structured end-to-end data science pipeline, progressing from raw data ingestion through to interactive business intelligence output. The architecture can be described in five sequential phases:

#	Phase	Description
1	Data Ingestion	Raw CSV file (Banking.csv) containing 3,000 records is loaded into a Pandas DataFrame using <code>pd.read_csv()</code> with appropriate delimiter and encoding parameters.
2	Data Preprocessing	Column standardization, removal of administrative identifiers, date type conversion, and validation of data completeness. Output is a clean analytical DataFrame.
3	Feature Engineering	Derivation of <code>Income_Band</code> , <code>Age_Group</code> , and <code>Total_Balance</code> from existing columns using <code>pd.cut()</code> and arithmetic operations. <code>GenderId</code> encoding mapped to binary values.
4	Exploratory Data Analysis	Statistical summaries and five key visualizations (scatter plot, bar chart, heatmap, distribution charts) generated using Seaborn and Matplotlib to surface patterns and relationships.
5	Machine Learning	K-Means clustering ($k=4$) applied to standardized features. Elbow Method used for k selection. Cluster labels assigned and segment names mapped for business interpretation.
6	Dashboard Integration	Segmented dataset exported and imported into Power BI Desktop. DAX measures created for KPIs. Six-page interactive dashboard built with cross-filtering, navigation, and drill-through capabilities.

The pipeline is designed for reproducibility — the Python analysis is contained in a Jupyter Notebook (Banking.ipynb), and the Power BI file (.pbix) references the output dataset. Re-running the notebook with updated data would automatically refresh all dashboard visuals upon reconnection.

9. Tools & Technologies

9.1 Python Libraries

Library	Version	Purpose
Pandas	Latest stable	Data loading, cleaning, transformation, and feature engineering
NumPy	Latest stable	Numerical operations and array manipulation
Seaborn	Latest stable	Statistical data visualization (heatmaps, scatter plots, bar charts)
Matplotlib	Latest stable	Plot rendering and figure customization
Scikit-learn	Latest stable	K-Means clustering, StandardScaler for feature normalization

9.2 Business Intelligence & Reporting

- Microsoft Power BI Desktop — Multi-page interactive dashboard with DAX measures, slicers, custom visuals, and navigation
- Jupyter Notebook — Integrated development environment for Python analysis and documentation

9.3 Development Environment

- Python 3.x — Core programming language for data analysis and ML
- Anaconda / pip — Package management and environment configuration
- CSV / XLSX — Data interchange formats between Python and Power BI

10. Key Business Insights

The analysis generated several actionable business insights with direct applicability to banking strategy and operations:

Insight Area	Finding & Recommendation
Income & Loan Relationship	Loan amounts show only weak correlation with income ($r \approx 0.33$). Banks should incorporate behavioral and credit history data into loan assessment models rather than relying primarily on declared income.
Deposit-Driven Customer Value	Total Balance is almost entirely driven by Bank Deposits ($r = 0.99$). The VIP segment (10.17% of customers) holds 34.39% of all deposits. These customers represent disproportionate revenue value and warrant dedicated relationship management.
Low-Income Risk Profile	Low-income customers exhibit the highest loan-to-income ratios relative to repayment capacity. Targeted risk monitoring and conservative credit limits are recommended for this segment.
Cross-Selling Opportunities	Strong correlations among banking products (deposits, checking, savings, foreign currency) indicate customers already engaged with multiple products. This presents opportunities for structured bundling and loyalty programs.
Customer Segmentation Value	Four distinct behavioral segments enable personalized outreach: VIP customers for wealth management products, High Risk customers for debt restructuring programs, Conservative customers for premium savings products, and Low Value customers for engagement activation campaigns.
Gender & Age Neutrality	Age and gender show minimal correlation with financial behavior, confirming that financial profile is a stronger segmentation basis than demographic profiling alone.

11. Conclusion

This project successfully demonstrates the application of data analytics and machine learning techniques to a real-world banking customer dataset. By constructing a comprehensive end-to-end pipeline — spanning data ingestion, preprocessing, feature engineering, exploratory analysis, unsupervised machine learning, and interactive visualization — the project delivers both technical rigor and business relevance.

The K-Means clustering model, validated through the Elbow Method, identified four coherent customer segments that align with recognizable banking archetypes: VIP depositors, active high-risk borrowers, conservative high-income customers, and low-engagement low-value accounts. These segments provide the foundation for differentiated marketing, risk management, and product personalization strategies.

The six-page Power BI dashboard translates the technical findings into an accessible, interactive format suitable for non-technical stakeholders including relationship managers, risk officers, and executive leadership. The dashboard's filtering capabilities enable drill-down analysis by income band, gender, segment, and date range, supporting dynamic decision-making at multiple levels of granularity.

Key findings confirm that deposit behavior is the most reliable indicator of customer value, that low-income customers represent elevated credit risk, and that cross-selling opportunities exist across account types. These insights are actionable and can be directly incorporated into bank strategy, underwriting models, and customer success programs.

12. Future Scope

While the current project delivers a robust analytics foundation, several enhancements could extend its capability and impact:

Enhancement	Description
Loan Default Prediction	Implement supervised ML models (Logistic Regression, Random Forest, XGBoost) trained on historical repayment data to predict probability of loan default. This would enable proactive risk management at the individual customer level.
Real-Time Dashboard Integration	Connect Power BI to a live database (PostgreSQL, Azure SQL) or streaming data source (Kafka, Event Hub) to enable real-time KPI monitoring and alert generation for risk thresholds.
Advanced Clustering Algorithms	Evaluate DBSCAN (density-based) and Hierarchical Agglomerative Clustering as alternatives to K-Means. These algorithms can identify clusters of arbitrary shape and do not require pre-specifying k.
Customer Lifetime Value (CLV) Modeling	Develop a CLV model to quantify the projected long-term revenue contribution of each customer segment, enabling more precise resource allocation in marketing and relationship management.
Natural Language Processing (NLP)	Incorporate customer feedback, complaint logs, and communication data using sentiment analysis to enrich the behavioral segmentation with attitudinal dimensions.
Power BI Service Deployment	Publish the dashboard to Power BI Service (cloud) to enable web-based access, scheduled data refresh, row-level security, and collaboration through shared workspaces.
AutoML Integration	Leverage automated machine learning platforms (AutoML, H2O.ai) for systematic model selection and hyperparameter tuning across multiple algorithms.

These enhancements would evolve the project from a descriptive analytics tool into a predictive and prescriptive intelligence platform — supporting not just understanding of past and current customer behavior, but proactive financial strategy and risk management.